ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 g	Conservation of mass.
2	**(b)** $Al_2(SO_4)_3$	Al3+ and SO4 2-.
3	**(b)** Mg and Cl	Metal and non-metal.
4	**(b)** 28 g	0.8235 x 34.
5	**(b)** $(NH_4)_3PO_4$	NH4+ and PO4 3-.
6	**(b)** copper	Metals have free electrons.
7	**(b)** 30 g	20 x 60/40.
8	**(b)** $Ca(OH)_2$	Ca2+ and OH-.
9	**(b)** O_2	Two shared pairs.
10	**(b)** 12 g	9 x 4/3.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mass of water = 18 g (1); Reactants = 18 g, products = 18 g (1).	1+1
12	H has valency 1 and O has valency 2 (1); Two H atoms are needed for one O atom: H_2O (1).	1+1
13	12/44 x 100 (1); = 27.3 per cent (1).	1+1
14	H : O = 1 : 8 (1); H = 4 g, O = 32 g (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	C = 3/11 x 55 = 15 g (1); O = 8/11 x 55 = 40 g (1); Total = 55 g (1).	1+1+1
16	NaCl ionic, CCl_4 covalent (1); NaCl high melting, CCl_4 low (1); NaCl conducts in solution, CCl_4 does not (1).	1+1+1
17	O is 30 per cent = 48 u (1); 2X = 70/30 x 48 = 112 (1); X = 56 u (iron) (1).	1+1+1
	OR: N = 6.6 - 4.8 = 1.8 g (1); Ratio = 4.8 : 1.8 (1); = 8 : 3 (1).	1+1+1
18	Ions are fixed in the solid (1); On melting the ions become free (1); Free ions carry charge (1).	1+1+1
	OR: CaCO3 gives CaO + CO2 (1); Mass of CaO = 200 - 88 (1); = 112 g (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ratio H : O in the first sample = 1 : 8 (1); Ratio in the second sample = 2 : 16 = 1 : 8 (1); The ratio is the same for different samples (1); Law: a pure compound always contains the same elements in a fixed ratio by mass (1); Proust gave this law; it distinguishes a compound from a mixture (1).	1+1+1+1+1
	OR: O = 1.5 g (1); Cu : O = 6 : 1.5 = 4 : 1 (1); CuO ratio = 63.5 : 16 = 3.97 : 1 (1); Agrees within error (1); Compound confirmed as CuO (1).	1+1+1+1+1
20	Na: 2, 8, 1 loses 1 (1); Mg loses 2, O gains 2: MgO (1); Ca loses 2; each F gains 1: CaF2 (1); Noble gas arrangement (1); Formulae by balance of charges (1).	1+1+1+1+1
	OR: Compare masses combined with a fixed mass of another element (1); Hydrogen was the first standard (1); Oxygen later (1); Carbon-12 is the present standard (1); u = 1/12 of the mass of a C-12 atom (1).	1+1+1+1+1

Mistakes to avoid:

- Treating the subscript in a formula as the valency, or forgetting to reduce criss-crossed numbers to the simplest ratio (writing Ca_2O_2 for calcium oxide).
- Writing FeO_3 or Al_3O_2 by swapping the wrong numbers; remember to criss-cross *valencies* (charges) and put the cation first.
- Forgetting the bracket when a polyatomic ion occurs more than once: $Ca(OH)_2$, not $CaOH_2$.
- Mixing atomicity with number of molecules: $3\,O_2$ means three molecules, six atoms; O_3 is one molecule of ozone.
- Adding the atomic mass of an element only once when the formula has a subscript or bracket (e.g. $(NH_4)_2SO_4$: N counted twice, H eight times).

- Calling NaCl a 'molecule' and giving it a 'molecular mass'; for ionic compounds use formula unit mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Fe_2O_3	Fe3+ and O2-.
2	**(b)** N_2	Three pairs.
3	**(b)** 22 g	6 + 16.
4	**(c)** $CuSO_4$	Cu2+ and SO4 2-.
5	**(c)** 4	Two double bonds.
6	**(d)** 3 : 8	12 : 32 = 3 : 8.
7	**(d)** 106 u	46 + 12 + 48.
8	**(d)** 2	Mg loses 2.
9	**(d)** 117 g NaCl	46 + 71.
10	**(a)** 111 u	40 + 71.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ca 1, O 2, H 2 (1); Total = 5 (1).	1+1
12	NaCl has strong ionic bonds in a lattice (1); HCl has weak forces between molecules (1).	1+1
13	24 + 2 x 17 (1); = 58 u (1).	1+1
14	Atoms are divisible into subatomic particles (1); Isotopes of an element have different masses (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mg loses two electrons: Mg^{2+} (1); Two Cl atoms each gain one: Cl^- (1); $MgCl_2$ formed (1).	1+1+1
16	Al3+ and SO4 2- (1); Cross the numerical charges (1); $Al_2(SO_4)_3$ (1).	1+1+1
17	Na: 2, 8, 1 loses 1 electron to form Na^+ (1); F: 2, 7 gains 1 electron to form F^- (1); Both attain a noble gas arrangement (1).	1+1+1
	OR: O has 6 valence electrons (1); Shares one electron with each of two H atoms (1); Two single bonds, octet on O, duet on each H (1).	1+1+1
18	Sum of atomic masses in the formula unit (1); 40 + 12 + 48 (1); = 100 u (1).	1+1+1
	OR: Giant covalent network (1); Each atom bonded to four others (1); Many strong bonds must break (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	C forms four single bonds in CH_4 (1); N forms three in NH_3 (1); O forms two in H_2O (1); C forms two double bonds in CO_2 (1); Octet satisfied (1).	1+1+1+1+1
	OR: Alternate Na+ and Cl- in a 3-D arrangement (1); Each ion surrounded by 6 oppositely charged ions (1); Strong electrostatic forces: high melting point (1); Brittle (1); Conducts when molten (1).	1+1+1+1+1
20	Melting points (1); Conductivity (1); Solubility (1); Hardness and brittleness (1); Reasons: strong lattice vs weak forces (1).	1+1+1+1+1
	OR: MgO 40, Mg 60 per cent (1); CaO 56, Ca 71.4 per cent (1); Fe2O3 160, Fe 70 per cent (1); CaO is highest (1); Atomic mass vs oxygen share (1).	1+1+1+1+1

Mistakes to avoid:

- Treating the subscript in a formula as the valency, or forgetting to reduce criss-crossed numbers to the simplest ratio (writing Ca_2O_2 for calcium oxide).
- Writing FeO_3 or Al_3O_2 by swapping the wrong numbers; remember to criss-cross *valencies* (charges) and put the cation first.
- Forgetting the bracket when a polyatomic ion occurs more than once: $Ca(OH)_2$, not $CaOH_2$.
- Mixing atomicity with number of molecules: $3\,O_2$ means three molecules, six atoms; O_3 is one molecule of ozone.

- Adding the atomic mass of an element only once when the formula has a subscript or bracket (e.g. $(NH_4)_2SO_4$: N counted twice, H eight times).
- Calling $NaCl$ a 'molecule' and giving it a 'molecular mass'; for ionic compounds use formula unit mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The Atomic Foundations of Matter** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 6	2 Al lose 6; 3 O gain 6.
2	**(b)** 3.2 g	8 - 4.8.
3	**(a)** 174 u	78 + 32 + 64.
4	**(c)** ions are free to move	Ions in solution.
5	**(a)** 11.1 per cent	2/18.
6	**(c)** 102 u	54 + 48.
7	**(c)** CH_4	Weak intermolecular forces.
8	**(b)** 1/12 of the mass of one carbon-12 atom	Definition of u.
9	**(a)** 180 u	72 + 12 + 96.
10	**(b)** repulsion between like ions when layers shift	Lattice shifting.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Molecule: discrete group of atoms (1); Formula unit: simplest ratio of ions in a lattice (1).	1+1
12	(a) KNO_3 (1); (b) $ZnSO_4$ (1).	1+1
13	Gas escapes (1); Mass of products is not fully measured (1).	1+1
14	(a) Calcium and chloride ions (1); (b) Sodium and sulphate ions (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	30 u (1); 28 u (1); 26 u (1).	1+1+1
16	H : O = 1 : 8 (1); 2 g H needs 16 g O (1); 16 g O is left unreacted (1).	1+1+1
17	Molecular mass for covalent molecules e.g. H_2O (1); Formula unit mass for ionic e.g. $NaCl$ (1); Both are sums of atomic masses (1).	1+1+1
	OR: C = 72 u in 180, so 6 atoms (1); H = 12 u, so 12 atoms (1); O = 96 u, so 6 atoms: $C_6H_{12}O_6$ (1).	1+1+1
18	C : O atoms 1 : 1 in CO (1); 1 : 2 in CO2 (1); Whole numbers (1).	1+1+1
	OR: Gas CO2 is trapped in the container (1); Total mass before and after is equal (1); In an open flask CO2 escapes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	NaNO3 mass = 85 (1); Na 27.1, N 16.5, O 56.5 per cent (1); (NH4)2SO4 mass = 132 (1); N 21.2, H 6.1, S 24.2, O 48.5 per cent (1); Sum 100 in each (1).	1+1+1+1+1
	OR: HCl: one shared pair; Cl has three lone pairs (1); H_2O: two shared pairs; O has two lone pairs (1); NH_3: three shared pairs; N has one lone pair (1); Octet on Cl, O, N; duet on H (1); Covalent bonds in all three (1).	1+1+1+1+1
20	Pure water is a compound with fixed H : O (1); Sea water and river water are mixtures (1); Dissolved salts vary (1); Law of constant proportions applies only to the compound (1); Pure water has the same composition anywhere (1).	1+1+1+1+1
	OR: Closed flask with two solutions in separate containers (1); Weigh before mixing (1); Mix and weigh again (1); Equal masses (1); Errors: leakage of gas, buoyancy, balance sensitivity (1).	1+1+1+1+1

Mistakes to avoid:

- Treating the subscript in a formula as the valency, or forgetting to reduce criss-crossed numbers to the simplest ratio (writing Ca_2O_2 for calcium oxide).
- Writing FeO_3 or Al_3O_2 by swapping the wrong numbers; remember to criss-cross *valencies* (charges) and put the cation first.
- Forgetting the bracket when a polyatomic ion occurs more than once: $Ca(OH)_2$, not $CaOH_2$.
- Mixing atomicity with number of molecules: $3\,O_2$ means three molecules, six atoms; O_3 is one molecule of ozone.

- Adding the atomic mass of an element only once when the formula has a subscript or bracket (e.g. $(NH_4)_2SO_4$: N counted twice, H eight times).
- Calling $NaCl$ a 'molecule' and giving it a 'molecular mass'; for ionic compounds use formula unit mass.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND